Santander
Corporate & Investment Banking

Macro Research Brazil

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Special Report – Credit

Introducing Santander's Credit Impulse Proxies

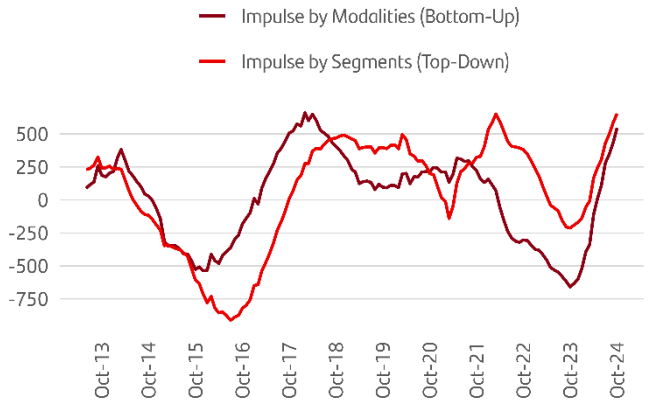
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Bottom line:

We are introducing two proxies for credit impulse: one from a top-down perspective and the other built from the bottom up. Regardless of the metric, we have found that currently the credit impulse is very high and has been driven by non-earmarked credit. Overall, the non-earmarked household credit impulse has been performing wel in 2024 — except for one modality, which saw a big spike in payments. In addition, the robust performance in the non-earmarked corporate credit impulse is also widespread, but payments have been prone to a substantial deceleration at the margin, which pushed the impulse even higher.

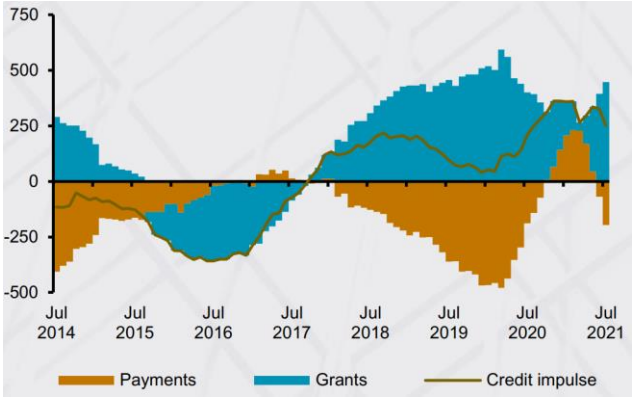
In this report, we explore two measures of credit impulse: one built from a top-down perspective, looking solely at the main segments of the bank credit market, with the other built from a bottom-up perspective, calculating the impulse of each modality of bank credit. While the first encompasses credit as a whole, the latter is limited to modalities for which complete data is publicly available but captures better actual credit payment dynamics.

Figure 1 – Santander's 12-month Credit Impulse Proxies
(BRL Bn, inflation-adjusted to current BRL)



Sources: BCB, Santander.

Figure 2 – BCB's 12-month Credit Impulse
(BRL Bn, inflation-adjusted to Jul-21 BRL)



Sources: BCB, Santander.

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Figure 1 shows our two proxies for credit impulse next to the one the Brazil Central Bank (BCB) presented in 2021[1], which Figure 2 illustrates. We note that both our proxies have a distinct shape from that of the BCB's. The reason behind these differences comes from the payments' series. The BCB does not disclose official bank credit payments' series. Thus, to present our proxies, we had to estimate payments. For our top-down approach, we estimated payments using aggregate data for each of the four main segments of bank credit: (i) non-earmarked household; (ii) non-earmarked corporate;,(iii) earmarked household; and (iv) earmarked corporate. For the bottom-up approach, we estimated payments for each modality of the main segments, which are displayed in Figure 3. We note that the listings in Figure 3 are non-exhaustive of all modalities. That is because some modalities do not have all the necessary data available for the estimation procedure and, thus, they had to be disregarded. We believe the final effect of not including them in the analysis to be limited since they are not modalities marked by large outstanding credit or new loans figures.

Figure 3 – Bank Credit Modalities Used

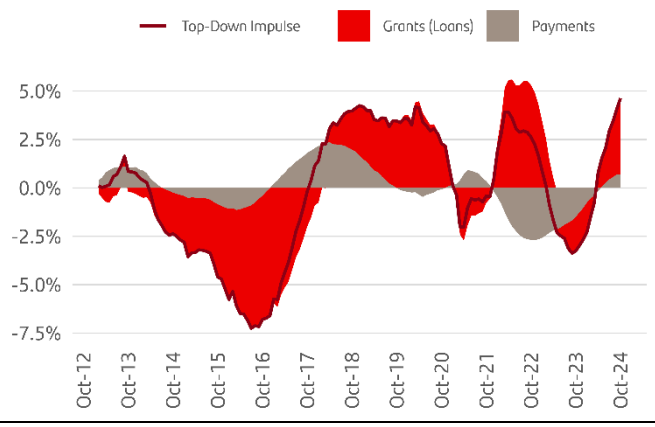
Nonearmarked		Earmarked	
Household	**Corporate**	**Household**	**Corporate**
Overdraft	Discount of trade bills and credit receivables	Rural credit - Market rates	Rural credit - Market rates
Personal credit - Excluding payroll deducted	Discount of checks	Rural credit - Earmarked rates	Rural credit - Earmarked rates
Personal credit - Payroll deducted - Public sector employees	Discount of credit card bills	Real estate financing - Market rates	Real estate financing - Market rates
Personal credit - Payroll deducted - Private sector employees	Working capital - Up to 365 days	Real estate financing - Earmarked rates	Real estate financing - Earmarked rates
Personal credit - Payroll deducted - Retirees and pensioners	Working capital - Over 365 days	BNDES funds - Investments	BNDES funds - Working Capital
Vehicles financing	Working capital - Revolving credit	BNDES funds - Agroindustrial	BNDES funds - Investments
Other goods financing	Guaranteed overdraft accounts	Microcredit - Consumption	BNDES funds - Agroindustrial
Credit card - Revolving	Overdraft	Microcredit - Microentrepreneurs	
Credit card - Financing	Vehicles financing		
Credit card - Non-financing	Other goods financing		
Leasing - Vehicles	Leasing - Vehicles		
Leasing - Other goods	Leasing - Other goods		
Discount of checks	Vendor		
	Compror		
	Credit card - Revolving		
	Credit card - Financing		
	Credit card - Non financing		
	Advances on exchange contracts		
	Imports financing		
	Exports financing		
	Foreign on lendings		

Sources: BCB, Santander.

Next, in Figures 4 and 5 we present the loans-payments breakdown of each of our impulse proxies. This set of pictures make clear the source of differences between methodologies. The top-down procedure estimates a very smooth payments' series, while the bottom-up one is far more volatile. We believe both series serve as complements: the first is complete in the sense that it considers all credit modalities when looking at segment aggregates, while the latter can deliver more realistic payment dynamics.

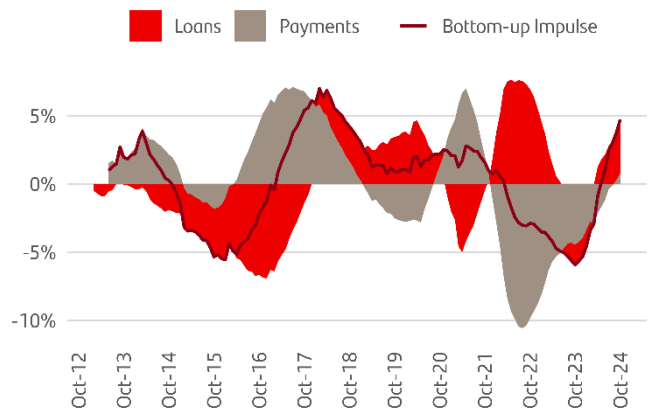
[1] To calculate the credit impulse, we follow the methodology proposed in BCB Inflation Report 3Q21 - Financial flow and credit stimulus.

Figure 4 – Santander's 12-month Top-down Credit Impulse (% GDP)



Sources: BCB, Santander.

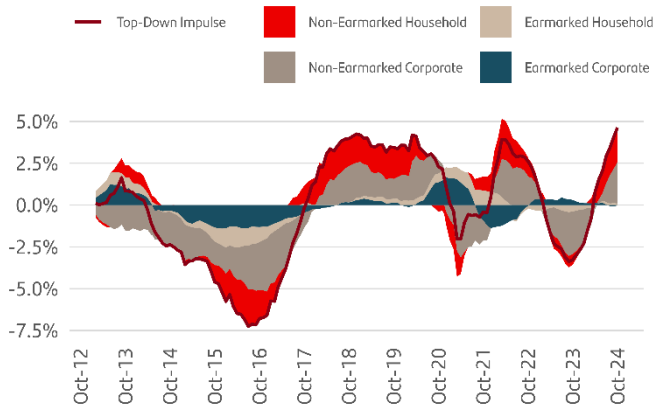
Figure 5 – Santander's 12-month Bottom-up Credit Impulse (% GDP)



Sources: BCB, Santander.

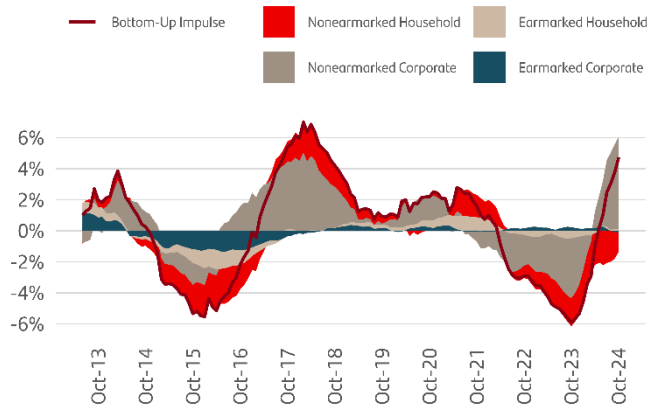
Figures 4 and 5 clearly demonstrate that despite the different methodologies, both impulse proxies indicate that currently the credit impulse is at approximately 4% of GDP. This finding is by itself of great relevance. At this point, we can detail a little further where this impulse comes from. Figures 6 and 7 break down each impulse at the segment level.

Figure 6 – Santander's 12-month Top-down Credit Impulse by Segments (% GDP)



Sources: BCB, Santander.

Figure 7 – Santander's 12-month Bottom-up Credit Impulse by Segments (% GDP)



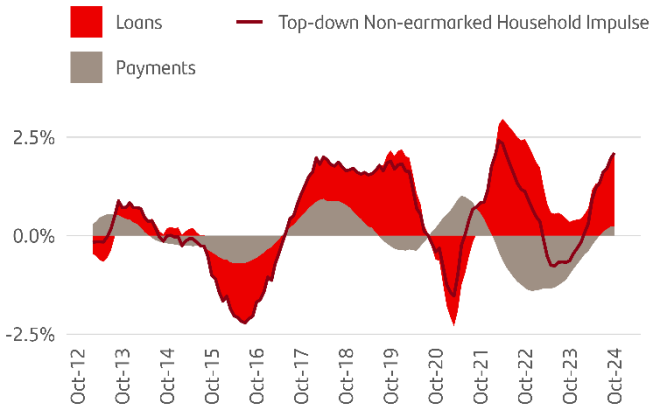
Sources: BCB, Santander.

Despite both impulse metrics being near each other at the margin, we reach strikingly different findings in terms of breakdown. Both metrics find that currently the earmarked credit impulse is approximately zero. But, **while the top-down approach indicates that the currently high impulse can be credited to non-earmarked credit to both households and corporations, the bottom-up impulse found a negative non-earmarked household impulse pushing down the non-earmarked corporate impulse, which is more than +5% of the GDP.**

Given the used methodologies, this difference must come from payments' series. Yet its sheer size calls for further investigation. Let's first look at the non-earmarked household part of each impulse metric. Figure 8 shows that the estimated payment series for non-earmarked household credit is currently delivering positive impulse, i.e., payments are falling for the segment, in a way that money

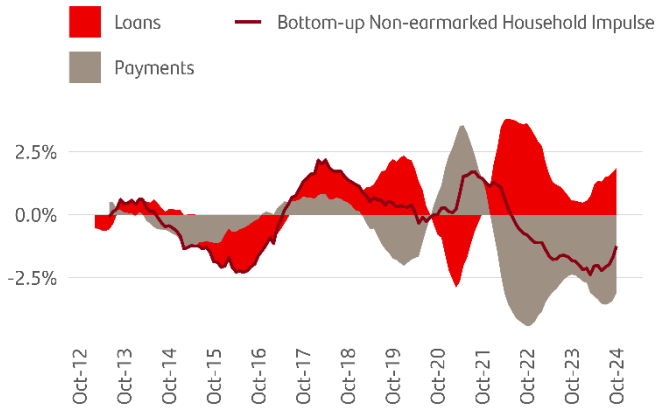
remains in the system (i.e., does not return to banks) and thus serve as an impulse. On the other hand, Figure 9 shows that this is not the case if you calculate payments from the bottom up. In fact, here payments are incredibly high and have been thus diminishing bank credit liquidity.

Figure 8 – Santander's 12-month Top-down Non-earmarked Household Credit Impulse (% GDP)



Sources: BCB, Santander.

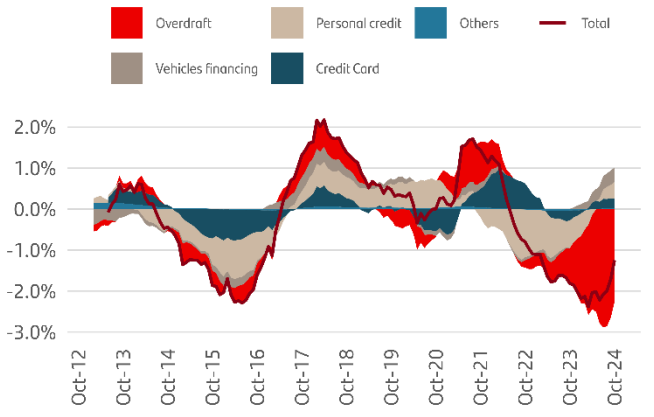
Figure 9 – Santander's 12-month Bottom-up Non-earmarked Household Credit Impulse (% GDP)



Sources: BCB, Santander.

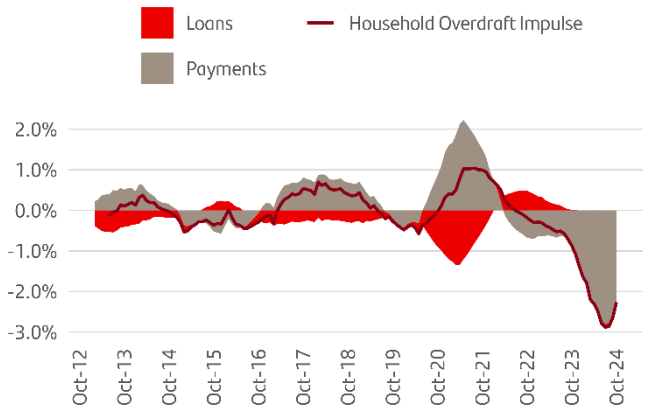
Even though we cannot go deeper in the top-down approach, we can take a look at which modality is driving the performance of the bottom-up non-earmarked household credit impulse. Figure 10 breaks down the impulse by modality. As we proceed with our investigation, the picture gets clearer and clearer. While indeed most non-earmarked household credit modalities have been delivering positive (and on aggregate large) impulse, overdrafts have been seeing a substantially large negative impulse. Figure 11 breaks down the impulse for this modality. It shows that since the start of the 2H23, payments accelerated substantially and thus drove this category's impulse down.

Figure 10 – Santander's 12-month Bottom-up Non-earmarked Household Credit Impulse by Modality (% GDP)



Sources: BCB, Santander.

Figure 11 – Santander's 12-month Bottom-up Non-earmarked Household Credit Impulse – Overdraft Breakdown (% GDP)

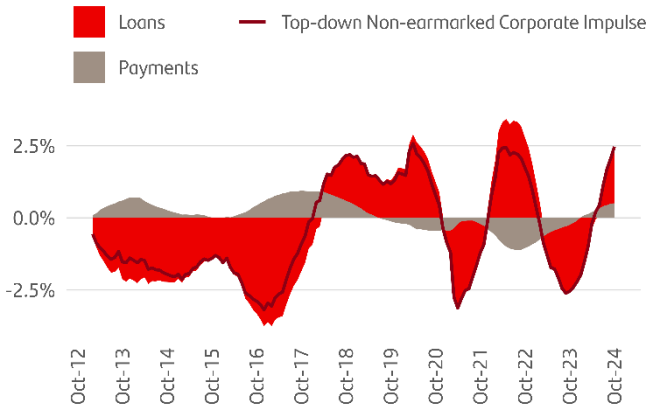


Sources: BCB, Santander.

We can now proceed to a similar investigation of the non-earmarked *corporate* impulse. Going back to figures 6 and 7, we have found that while the top-down approach calculated a 2% of GDP impulse from this category, the bottom-down approach found it above 5%

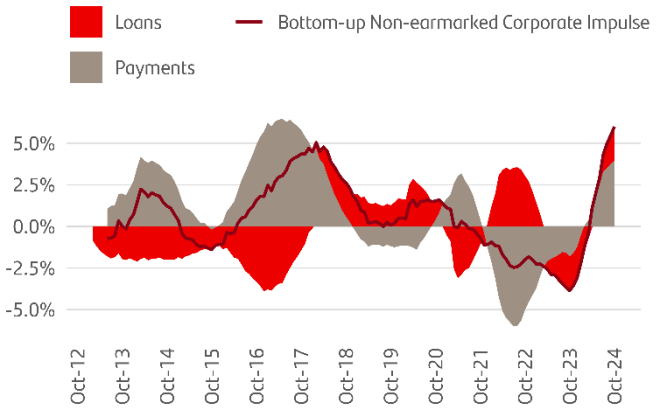
of GDP. We break down this segment's impulse in Figures 12 and 13. Once again, we find excessively smooth payments in the top-down approach, which explains the difference between both series. Figure 14 complements the analysis and shows that the corporate impulse is largely positive among major categories, but the highlight is on the discount of credit card bills. Given the strong competition in this modality, it is possible that the definition of pre-negotiated conditions for automatic and immediate credit has made this modality even more dependent on retail and services consumption. In other words, its pro-cyclicality has likely increased. Figure 15 shows both new loans and lack of payments are driving the impulse for this modality, which by itself adds to ~4% GDP.

Figure 12 – Santander's 12-month Top-down Non-earmarked Corporate Credit Impulse (% GDP)



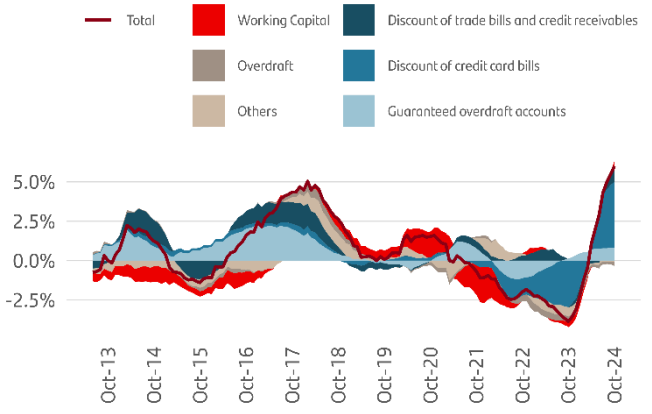
Sources: BCB, Santander.

Figure 13 – Santander's 12-month Bottom-up Non-earmarked Corporate Credit Impulse (% GDP)



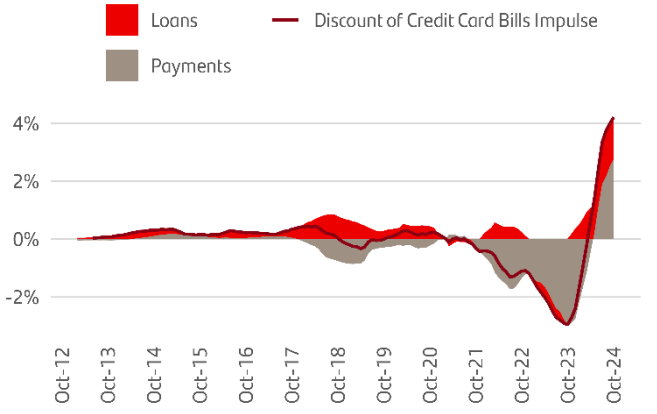
Sources: BCB, Santander.

Figure 14 – Santander's 12-month Bottom-up Non-earmarked Corporate Credit Impulse by Modality (% GDP)



Sources: BCB, Santander.

Figure 15 – Santander's 12-month Bottom-up Non-earmarked Corporate Credit Impulse – Discount of Credit Card Bills (% GDP)



Sources: BCB, Santander.

In summary, we have found that currently the credit impulse is very high. Looking at segments' aggregates might lead to distorted conclusions when payments for specific modalities are prone to large and volatile changes.

Overall, the non-earmarked household credit impulse has performed well in 2024 — except for one modality (overdrafts), which saw a big spike in payments. Despite distorting the aggregate figure negatively, the composition of the non-earmarked household impulse is good, in our view: the negative impulse comes from an acceleration in payments in a modality marked by elevated risk.

In addition, the robust performance of the non-earmarked corporate impulse is widespread, but a single modality (discount of credit card bills) leads the race by far. This composition is not the best, since part of this impulse comes from a lack of payments, and this modality's credit is traditionally used for short-term corporate activities, and not long-term investments.

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